📄 Task 3.1: Network Concepts

1. OSI Model (3 points)

7 Layers of OSI Model

1. **Physical Layer**
2. **Data Link Layer**
3. **Network Layer**
4. **Transport Layer**
5. **Session Layer**
6. **Presentation Layer**
7. **Application Layer**

Example Protocols for Each Layer

Layer	Name	Example Protocol
7	Application	HTTP, FTP
6	Presentation	SSL/TLS
5	Session	NetBIOS
4	Transport	TCP, UDP
3	Network	IP
2	Data Link	Ethernet
1	Physical	Cables, Signals

Which Layer TCP Operates At

- **TCP operates at the Transport Layer (Layer 4)**
- It ensures **reliable data transfer**, error checking, and flow control.

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2. IP Addressing (3 points)

Difference Between Public and Private IP

Public IP	Private IP
Used on the internet	Used inside local network
Globally unique	Not unique (can repeat)
Assigned by ISP	Assigned by router
Accessible from anywhere	Not directly accessible

Private IP Ranges

- **Class A:** 10.0.0.0 – 10.255.255.255
- **Class B:** 172.16.0.0 – 172.31.255.255
- **Class C:** 192.168.0.0 – 192.168.255.255

Is 192.168.1.100 Public or Private?

- **192.168.1.100 is a Private IP address**
- It belongs to the **Class C private range**

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3. Protocols (4 points)

Port Numbers and Services

Port	Service
22	SSH (Secure Shell)
80	HTTP (Web)
443	HTTPS (Secure Web)
3306	MySQL Database

Difference Between TCP and UDP

TCP	UDP
Connection-oriented	Connectionless
Reliable (guarantees delivery)	Unreliable
Slower	Faster
Error checking	No error checking
Used in web, email	Used in streaming, gaming

What is NAT and Why is it Needed

NAT (Network Address Translation) is a method used by routers to:

- Convert **private IP addresses → public IP address**
- Allow multiple devices to share one public IP

Why NAT is Needed

- Saves IPv4 addresses
- Improves security (hides internal network)
- Allows internet access for private devices